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Prof. Diane Joseph-McCarthy, PhD, Editor-in-Chief
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Dear Prof. Joseph-McCarthy,

We submit our manuscript “**Do topological and quantum-inspired molecular representations add value? Evidence from African antimalarial chemical space**” for consideration as an Original Research Article in *Journal of Computer-Aided Molecular Design*.

This work addresses a computational method evaluation question relevant to computer-aided drug design: whether physics-inspired molecular representations—persistent homology-based topological fingerprints, tensor network compression, and quantum kernel methods—offer practical advantages over established classical fingerprints for virtual screening and molecular property prediction. We present a rigorous benchmarking study comparing three novel descriptors (TFP, TNE, QKS) against ECFP4 and other classical baselines on 19 849 African natural-product-derived antimalarial candidates, with computational activity labels from the Ersilia eos80ch model. The evaluation uses matched 5-fold cross-validation protocols, ablation analysis, and includes a label-source-shift transfer assessment on 22 447 ChEMBL compounds.

Key methodological contributions and findings:

1. **Topological descriptor for scaffold analysis:** persistent homology decomposes the scaffold expansion paradox (92.6% whole-molecule Tanimoto-distinct yet 69.3% scaffold-conserving) into H_0 (divergent connectivity) versus H_1 (conserved ring topology). TFP achieves cross-validated AUC 0.876, providing structural interpretability but not predictive superiority over ECFP4 (0.948).
2. **Tensor network molecular compression:** Tucker decomposition at bond dimension $d = 8$ achieves $6.1\times$ real-atom compression (mean 39 atoms $\rightarrow$ 192 elements; reconstruction error 0.113). TNE retains docking-score-predictive information on PfDHFR ($R^2 = 0.473$ versus ECFP4 0.461) but degrades on cross-validated activity prediction (AUC 0.722).
3. **Quantum kernel baseline comparison:** simulated 6-qubit quantum kernels showed no statistically detectable difference from properly tuned RBF kernels internally (AUC 0.823 versus 0.829, $p = 0.060$), and performed lower on ChEMBL transfer (AUC 0.817 versus 0.847, $p = 0.021$). This honest-negative result establishes that current quantum kernel implementations do not outperform optimized classical baselines for this application.
4. **Hybrid descriptor evaluation:** the combined TFP+TNE+QK descriptor achieves AUC 0.888, below ECFP4 (0.948, $p < 0.0001$). Ablation analysis identifies QKS as the principal contributor ($\Delta = -0.040$), while stacking TFP and TNE onto ECFP4 yields no detectable gain ($\Delta\text{AUC} = -0.0002$, $p = 0.084$).
5. **Scaffold-based validation:** Bemis–Murcko scaffold-grouped cross-validation (panel spans 632 scaffolds) reveals all descriptors lose 0.086–0.196 AUC, with ECFP4 dropping from 0.948 to 0.822, quantifying scaffold memorization effects across descriptor classes.
6. **Computational efficiency and practical applicability:** all three methods are implemented with open-source libraries (Ripser.py, TensorLy, PennyLane) and scale to tens of thousands of molecules. TNE compression reduces storage requirements while TFP provides interpretable topological features for post-hoc analysis. Complete computational protocols are provided for reproducibility.

Scope alignment with JCAMD: this work directly addresses the journal’s focus on computational method development and rigorous benchmarking for drug discovery applications. The study provides: (i) three reusable molecular descriptor implementations with full algorithmic details, (ii) controlled baseline comparisons under matched cross-validation protocols, (iii) explicit performance boundaries through scaffold-based validation and external transfer assessment, (iv) honest reporting of negative findings where novel methods did not outperform classical approaches, and (v) practical guidance on when these methods add value (structural interpretation, compression) versus when they do not (primary virtual screening).

Evidence boundaries: all activity labels are computational predictions (Ersilia eos80ch model), not experimental measurements. The ChEMBL transfer panel is label-source-independent but not confirmed molecule-level-disjoint from the internal panel, representing a transferability assessment rather than strict external validation. The quantum kernel outperformed linear kernels but not the gamma-tuned RBF baseline, establishing that careful classical method optimization remains competitive.

Reproducibility: complete computational provenance (persistent homology calculations, Tucker decomposition parameters, quantum kernel circuits, cross-validation splits, ablation protocols, baseline hyperparameter grids) is archived at Zenodo (<https://doi.org/10.5281/zenodo.22768267>) under CC BY 4.0 license. The Supporting Information (16 pages) contains implementation pseudocode, descriptor generation workflows, and complete benchmark tables. All random seeds are fixed (`random_state=42`) for exact reproducibility.

This manuscript is original, not under consideration elsewhere, and all authors have reviewed and approved the submission. We declare no competing financial interests.

Sincerely,

Myke Vital Sao Temgoua (on behalf of all co-authors)